3.9 Specific landslide mitigation measures for the study areas

3.9.1 Assumptions

This study has been carried based on literature review, satellite imagery review, and that no specific field measurements were taken; therefore the determination of landslide mitigation measures appropriate for each site has been based on assumptions. From these, cost estimates were produced.

The mitigation measures hereafter presented are proposed for areas in each sub-catchment with the highest susceptibility to landslides, i.e. landslide hazard classes of 365 m^2/year/km^2 and 90m^2/year/km^2. Section 1 provides details of the landslide risk assessment.

The total surface areas covered by the two landslides hazard classes are given in the table below.

Table 15: **Surface areas under the two highest landslide hazard classes**

Sub-catchment	Total surface area (ha)	Combined area under the landslide hazard classes of 365 m^2/year/km^2 and 90m^2/year/km^2
Bishenyi	4,686.8	666.7
Rwabayanga	809.0	68.4
Rwandex – Magerwa	979.3	63.9
Kamembe – Gihundwe	2,123.6	76.3

The following assumptions are made:

- Should landslides occur in the two aforementioned highest susceptibility zones, they would be shallow landslides that are likely to cause soil erosion;
- Given the assumption of shallow landslides, mitigation measures are designed to address the following physical processes:
    - Surface protection and control of erosion;
    - Modification of land geometry
    - Modification of the surface water regime
- The mitigation measures proposed will not include elements that lead to an increase in surface water infiltration into the soil. It is has been discussed in the preceding section that any soil erosion control measure that leads to increased water infiltration (e.g. radical terraces) would lead to creation of conditions that are favourable to landslides;
- Within each area with the highest susceptibility to landslides, it is assumed that the total coverage of zones to be reforested is 30%. This assumption builds from the guidance of the Rwanda building code that stipulates a coverage of 30% for landscaping (trees, green spaces) in urban developments;
- For drainage trenches / ditches, a coverage of 500 m of channel per 3 ha is assumed.